

CSE 202 Midterm 3 Part (b)

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I. MULTIPLE CHOICE QUESTIONS (40 POINTS)

1. Given the following data $a = (16-14-10-8-7-9)$ as a max-heap. What will be the heap after the following operations?

```
a.deleteMax();
a.deleteMax();
a.insert(12);
a.insert(3);
```

- a) 12-10-9-8-7-3
- b) 12-10-9-3-8-7
- c) 12-10-7-9-8-3
- d) 12-10-7-8-9-3
- e) 12-9-10-8-3-7

2. What is the time complexity of search and deleteMax for the min-heap?

- a) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
- b) $\mathcal{O}(1)$, $\mathcal{O}(N)$
- c) $\mathcal{O}(N)$, $\mathcal{O}(\log N)$
- d) $\mathcal{O}(1)$, $\mathcal{O}(\log N)$
- e) $\mathcal{O}(\log N)$, $\mathcal{O}(1)$

3. If the height of the min-heap is h , what is the maximum number of elements that can we store in that heap?

- a) 2^h
- b) $2^h - 1$
- c) 2^{h-1}
- d) $2h - 1 + 1$
- e) $2^h + 1$

4. If the index of a node is t , what are the index of the grandparent of that node?

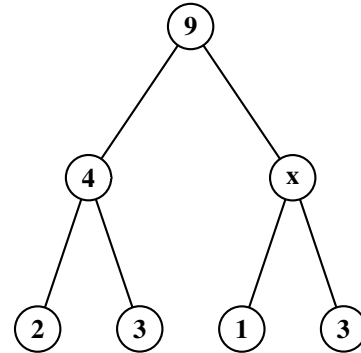
- a) $(t - 2) / 3$
- b) $(t - 2) / 4$
- c) $(t - 3) / 3$
- d) $(t - 3) / 4$
- e) $(t - 4) / 4$

5. If the height of the d-heap is h , what is the minimum number of elements that can we store in that heap?

- a) d^h
- b) $\frac{d^h - 1}{d - 1}$
- c) $\frac{d^h + d - 1}{d - 1}$
- d) $\frac{d^{h-1} + d - 2}{d - 1}$
- e) $d^h - 1$

6. For heap above, what can be the maximum and minimum value of x ?

- a) 7, 5
- b) 7, 4
- c) 4, 2
- d) 5, 2



- e) 8, 4

7. In the end of the disjoint set algorithm, if all sets are combined, what are the number of disjoint sets in terms of N (number of sets)?

- a) 1
- b) 2
- c) $N + 1$
- d) $2N$
- e) $2N + 1$

8. What is the time complexity of finding the (i) grandparent of a set, (ii) parent of a set, given the index of the set?

- a) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
- b) $\mathcal{O}(1)$, $\mathcal{O}(N)$
- c) $\mathcal{O}(N)$, $\mathcal{O}(1)$
- d) $\mathcal{O}(1)$, $\mathcal{O}(1)$
- e) $\mathcal{O}(\log N)$, $\mathcal{O}(1)$

9. If heights of two disjoint sets are 7 and 7, what is the height of the resulting set if those sets are merged?

- a) 5
- b) 6
- c) 7
- d) 8
- e) 9

10. Given a disjoint set of a contains 10 elements initially (0 to 9), what is the number of disjoint sets after the following operations?

```
a.union(0, 2);
a.union(1, 4);
a.union(2, 5);
a.union(4, 7);
a.union(0, 5);
a.union(1, 5)
```

- a) 3
- b) 4
- c) 5

- d) 6
e) 7

11. Given a disjoint set of N elements initially, what is the maximum number of representative sets?

- a) $N/2$
b) N
c) 2
d) $N - 1$
e) $N - 2$

12. Which of the following conditions will be true if a node with index i is not a representative set?

- a) `sets[i] == i`
b) `sets[i] != i`
c) `sets[i].parent == i`
d) `sets[i].parent != i`
e) `sets[i].parent.parent != i`

13. What is the maximum number of edges in an unweighted graph with N nodes?

- a) N
b) $N(N - 1)$
c) $\frac{N(N-1)}{2}$
d) N^2
e) $\frac{N(N+1)}{2}$

14. What is the minimum number of connected components of a graph with N nodes?

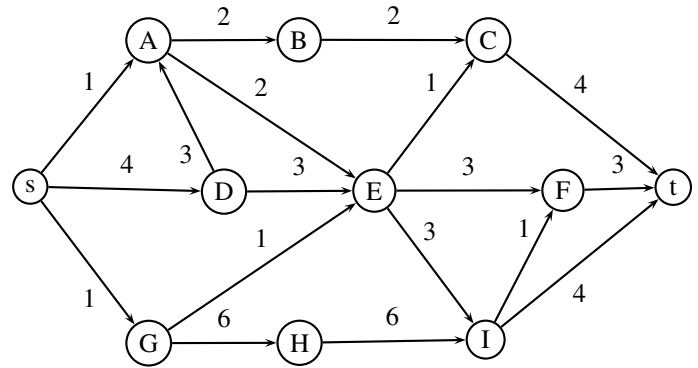
- a) $N / 4$
b) N
c) 2
d) $N - 1$
e) 1

15. What is the time complexity of the algorithm (for a graph with N nodes) which checks if there is an edge from node i to itself for (i) adjacency matrix, (ii) adjacency list representation?

- a) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
b) $\mathcal{O}(1)$, $\mathcal{O}(N)$
c) $\mathcal{O}(N)$, $\mathcal{O}(1)$
d) $\mathcal{O}(\log N)$, $\mathcal{O}(1)$
e) $\mathcal{O}(1)$, $\mathcal{O}(1)$

16. For an unweighted undirected graph G with N nodes and E edges, what is the sum of the elements of column i of the adjacency matrix of G ?

- a) E
b) number of incoming edges to i
c) N
d) number of outgoing edges to i
e) $2E$



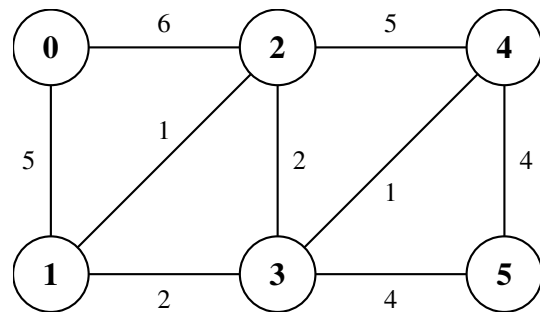
17. In the graph above, what is the shortest path from s to t ?

- a) 11
b) 10
c) 9
d) 8
e) 7

18. If you want to solve the all pairs source shortest path problem, which algorithm should you use?

- a) Dijkstra
b) Floyd-Warshall
c) Bellman-Ford
d) Kruskal
e) Prim

19. Given the graph below, what is the length of the minimum



spanning tree?

- a) 11
b) 12
c) 13
d) 14
e) 15

20. What is the worst case time complexity of Floyd-Warshall algorithm?

- a) N^4
b) N^2
c) $N^2 \log N$
d) $N^3 \log N$
e) N^3